Figure 5.3 Asymmetric key encryption and decryption

Thus, the public-private key pair has a unique property: when a message is encrypted with the private key, it can be decoded with the public key, and vice versa. The encryption and decryption keys are not the same; thus, the method is asymmetric. Now the key distribution problem is solved: you can publicize the public key for anyone to use, keeping the private key safe and secure. This property helps solve not only the key distribution issue, but also the issue of decentralized participant identity.

Next, let’s explore how public-key cryptography is used to address many issues in blockchain and decentralized applications.

5.2 The relevance of public-key cryptography to blockchain

Public-key cryptography is used for a range of operations on the blockchain, from account address generation to transaction signing.

5.2.1 Generating Ethereum addresses

As you learned in chapter 2 (section 2.5), there are two types of accounts in Ethereum: externally owned accounts (EOAs) and smart contract accounts. Open the Remix IDE, enter any smart contract, compile, and deploy. You’ll see the addresses in the left panel, as shown in figure 5.4.